Machine Learning Project Report

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Abstract - The project analysis of the Yellow Taxi Trip Records dataset from New York City's official taxis, for the given month, November 2025, is carried out. The purpose of the project is to find out the impact of the factors such as the distance of the trip, the number of passengers carried, the fare amount, and the mode of payment on the general functioning of the taxis. Through data pre-processing and the application of various data analysis techniques and simple statistical methods, the dataset was analyzed. Variations were found in the fare amount and the behavior of the passengers with respect to the mode of payment. The project analysis is useful and can be further developed into a predictive model. The data dictionary describes the structure of the data and the data used in this research.

Index Terms - Machine Learning, Data Analysis, Exploratory Data Analysis (EDA), Transportation Analytics, Fare Prediction, Feature Engineering, Trip Distance Analysis, Passenger Patterns, Data Preprocessing, Statistical Modeling

1. INTRODUCTION

Urban transportation systems are known to generate enormous data every day, which is an invaluable source of insights. Taxi services are an integral part of urban transportation systems. Therefore, analyzing data from taxi services is of significant importance. Yellow Taxi Trip Records is a dataset containing details of taxi trips. It includes details such as pick-up time, drop time, distance traveled, amount paid, number of passengers, and mode of payment. This data analysis is helpful in understanding how urban transportation systems function in real-world scenarios.

Problem Statement

The problem in dealing with large-scale data is extracting useful insights from raw data. In other words, it is challenging to process and analyze data. In this project, we are dealing with taxi trip data. We are expected to analyze this data and find useful insights in terms of patterns and correlations between key variables such as distance, amount paid, number of passengers, and mode of payment. In other words, we are expected to process raw data and extract useful information from it. This useful information can be used to explain how taxi services function in real-world scenarios.

Importance of the Topic

Data analysis of taxi trips is of significant importance. There are several reasons why taxi data analysis is of importance. First and foremost, it is helpful in understanding urban mobility. Urban mobility is an important aspect of urban planning. Therefore, analyzing data from taxi trips is of importance. In addition, it is helpful in understanding how to improve customer satisfaction. In other words, it is helpful in understanding how to improve taxi services. Furthermore, it is helpful in understanding passenger behavior. In other words, it is helpful in understanding how to improve taxi services. Therefore, it is of significant importance. In conclusion, it is of importance because it is helpful in understanding how data analysis techniques can be applied in real-world scenarios.

This report is specifically concerned with the analysis of the Yellow Taxi Trip Records for November, 2025. The data set contains relevant information pertaining to taxi trips, including time stamps for both pickup and drop-off, passenger count, trip distance, amount of fare, mode of payment, and other location identifiers. The data set provides an organized overview of real-world events in transportation. It is also backed by an in-depth data dictionary that provides definitions for all variables in the data set. The data set contains several features that capture both operational and financial aspects of taxi trips. Some of the variables in the data set include trip distance, total fare, tip amount, mode of payment, among others.

These variables are fundamental in capturing operational aspects of taxi trips. The project also involves several aspects of data analysis. To start with, the data set is preprocessed. This involves addressing missing values, correcting errors, and setting appropriate data types for each feature in the data set. The project then involves Exploratory Data Analysis (EDA), which is conducted to understand the distribution of variables in the data set. The project also involves

analyzing relationships between trip distance and fare, passenger count, and mode of payment. Visualization is also used in conjunction with data analysis. The project is concerned with generating insights from the data set. Therefore, it provides an in-depth understanding of how taxi services work. Based on the project, it is evident that data preprocessing and analysis techniques have been applied effectively in generating insights from the data set. The project is also clear on how data analysis is conducted.

2. Methodology

The methodology of this analysis follows a comprehensive, end-to-end data science pipeline. The system operates by ingesting raw NYC Yellow Taxi trip data, cleaning it, exploring statistical trends, preparing features, and ultimately training both predictive and unsupervised machine learning models. The step-by-step working of the system is detailed below.

2.1 Data Acquisition and Preprocessing

The system begins by loading a large-scale dataset (over 4.1 million records) stored in a Parquet file format using the **Pandas** library. To ensure the models receive high-quality data, rigorous **Data Cleaning** techniques are applied. Missing values are identified using .isnull() and handled appropriately. The system filters out anomalous and erroneous records through conditional subsetting, specifically removing instances with zero or null passengers, negative or zero fare amounts, highly unreasonable fares (e.g., >$500), negative tips, and impossible trip distances.

2.2 Exploratory Data Analysis (EDA) and Statistical Analysis

To understand the underlying patterns of taxi usage, the system utilizes **Matplotlib** and **Seaborn** for data visualization. **Histograms** are generated to observe the naturally right-skewed distribution of fare amounts, while **Boxplots** are implemented to detect outliers in numerical columns like trip distance and tips. Temporal trends are visualized using **Bar Charts** to group trips by hour of the day and day of the week.

Mathematically, the system calculates **Descriptive Statistics** including **Mean, Median, Standard Deviation**, as well as **Skewness and Kurtosis** using the scipy.stats module to quantify the asymmetry and tail-heaviness of the financial data. **Percentile Analysis** is also executed to understand the spread of fares. Finally, **Correlation Heatmaps** are computed using Pearson correlation to identify linear relationships between features (e.g., the high correlation between trip distance and fare amount).

2.3 Feature Engineering and Selection

Machine learning algorithms require uniformly scaled and appropriately formatted data to function efficiently. The system prepares the feature matrix through several steps:

- **Feature Scaling:** Numerical features are standardized using two techniques:

StandardScaler (which centers data to have a mean of 0 and a standard deviation of 1) and **MinMaxScaler** (which bounds data strictly between 0 and 1).

- **Categorical Encoding:** The system applies **Label Encoding** to transform ordinal vendor and rate IDs into integer formats. It applies **One-Hot Encoding** (via Pandas get_dummies) to nominal features like payment types, converting them into binary vector representations.
- **Feature Selection:** To improve computational efficiency and reduce noise, **Variance Thresholding** is implemented to drop features that show little to no variance. **Correlation-based Feature Selection** is also used to rank variables by their relationship to the target variable.

2.4 Predictive Modeling (Classification & Regression)

The system leverages the scikit-learn library to train predictive models, splitting the dataset into training (80%) and testing (20%) sets.

- **Classification (Predicting High Tips):** To predict whether a tip will be "high" (defined as above the median tip threshold), the system creates a binary target variable. It implements a **Logistic Regression** model for baseline linear probability estimation, alongside non-linear tree-based models: **Decision Tree Classifier** and **Random Forest Classifier** (an ensemble method using multiple decision trees to improve accuracy).
- **Regression (Predicting Total Fare):** To predict the exact total fare amount, the system implements standard **Linear Regression**, along with its regularized variants, **Ridge Regression** (L2 penalty) and **Lasso Regression** (L1 penalty), to prevent overfitting. Tree-based regressors (**Decision Tree Regressor** and **Random Forest Regressor**) are also deployed to capture complex, non-linear dependencies.

2.5 Model Evaluation and Hyperparameter Tuning

The models are rigorously evaluated using mathematical metrics:

- **Classification Metrics:** The system calculates **Accuracy, Precision, Recall, and F1-Score**. Visual tools like the **Confusion Matrix**, **ROC-AUC Curves**, and **Precision-Recall Curves** are generated to evaluate the trade-offs between true positive and false positive rates.
- **Regression Metrics:** Models are judged using **R-squared (R^2)**, **Root Mean Squared Error (RMSE)**, and **Mean Absolute Error (MAE)**. **Residual Analysis Scatter Plots** are used to verify that prediction errors are randomly distributed around zero.

To optimize the models, the system applies **K-Fold Cross-Validation** to ensure the algorithm generalizes well to unseen data. It searches for the best structural parameters (such as tree maximum depth and number of estimators) using **GridSearchCV** (an exhaustive search over a specified parameter grid) and **RandomizedSearchCV** (a probabilistic search). **Learning Curves** are plotted to diagnose if the model is suffering from high bias (underfitting) or high variance (overfitting).

2.6 Unsupervised Learning (Clustering)

To discover hidden segments within the taxi trips without predefined labels, the system utilizes unsupervised clustering algorithms:

- **K-Means Clustering:** Trips are grouped based on features like fare, distance, and tips. The system determines the optimal number of clusters (k) by implementing the **Elbow Method** (plotting inertia/within-cluster sum of squares) and calculating **Silhouette Scores** (measuring cluster cohesion versus separation).

- **Dimensionality Reduction:** To visualize these multi-dimensional clusters on a 2D graph, **Principal Component Analysis (PCA)** is applied to compress the feature space while retaining maximum variance.

- **DBSCAN (Density-Based Spatial Clustering of Applications with Noise):** This algorithm is implemented specifically for outlier detection. Unlike K-Means, DBSCAN groups dense core points and isolates sparse, anomalous trips (e.g., exceptionally long or expensive rides) as noise. The optimal density radius (eps) is calculated using a **K-Distance Graph** via the **NearestNeighbors** algorithm.

2.7 Natural Language Processing (NLP) and Text Analytics

The system includes an NLP module to process synthesized text descriptions of the taxi trips.

- **Text Preprocessing:** Using the **NLTK** library, raw text undergoes **Tokenization** (splitting strings into individual words) and **Stopword Removal** (filtering out non-informative words). The system contrasts **Stemming** (using a PorterStemmer to chop word endings) against **Lemmatization** (using a WordNetLemmatizer to reduce words to their proper dictionary root).

- **Text Vectorization:** To feed text into machine learning models, it is converted into numeric matrices using **Bag of Words (CountVectorizer)** and **TF-IDF (Term Frequency-Inverse Document Frequency)**, which mathematically down-weights overly common words.

- **Document Similarity & Clustering:** The textual similarity between different trip descriptions is calculated using a **Cosine Similarity Matrix**. Finally, **Document Clustering** is performed by applying K-Means to the TF-IDF vectors, effectively grouping trips by their textual profiles.

- **Generative AI Concepts:** A foundational **N-gram (Bigram) Model** is implemented to demonstrate generative text capabilities based on word sequence probabilities. This is accompanied by practical applications of Prompt Engineering techniques, such as **Zero-shot, Few-shot, and Chain-of-thought** prompting.

3. RESULTS AND OUTPUT

	Missing	Pct
passenger_count	1014740	24.27
RatecodeID	1014740	24.27
store_and_fwd_flag	1014740	24.27
congestion_surcharge	1014740	24.27
Airport_fee	1014740	24.27

Table 1: Missing Values Assessment. *Description:* A tabular breakdown of columns containing null values (e.g., passenger_count, RatecodeID) before data cleaning.

Inference: Approximately 24.27% of records lacked critical data. These were successfully dropped to ensure models were trained only on valid, complete trip records.

	VendorID	tpep_pickup_datetime	tpep_dropoff_datetime	passenger_count	trip_distance
count	3061480.00	3061480	3061480	3061480.00	3061480.00
mean	1.88	2025-11-15 10:40:21.022958	2025-11-15 10:58:30.810329	1.29	3.42
min	1.00	2008-12-31 23:04:21	2008-12-31 23:32:25	1.00	0.01
25%	2.00	2025-11-08 00:56:39	2025-11-08 01:10:52.750000	1.00	1.00
50%	2.00	2025-11-15 07:42:12.500000	2025-11-15 07:55:34.500000	1.00	1.70
75%	2.00	2025-11-22 06:35:54.250000	2025-11-22 06:53:03.750000	1.00	3.33
max	7.00	2025-11-30 23:59:59	2025-12-01 21:41:00	8.00	99.81
std	0.83	NaN	NaN	0.70	4.52

Table 2: Descriptive Statistics. *Description:* Summary statistics (mean, min, max, std, quartiles) for all numerical features after cleaning.

Inference: The average passenger count is 1.29, and the average trip distance is 3.42 miles. The extreme maximums (e.g., max distance 99.81 miles) highlight the necessity of the earlier outlier removal steps.

	Skewness	Kurtosis
fare_amount	3.349	25.796
trip_distance	2.894	12.021
tip_amount	5.810	326.038
total_amount	3.136	19.514
tolls_amount	5.512	56.345
passenger_count	3.002	10.448

Table 3: Distribution Metrics. *Description:* Tables calculating the skewness, kurtosis, and side-by-side mean/median/mode for fare-related columns.
Inference: fare_amount and tip_amount show high positive skewness (3.349 and 5.810) and massive kurtosis. The mean is consistently higher than the median, confirming heavy right-tailed distributions (pulled up by a few very expensive trips)

Fare Amount Percentiles:

	Percentile	Fare ($)
0	5	5.8
1	10	7.2
2	25	9.3
3	50	14.2
4	75	23.3
5	90	42.2
6	95	66.0
7	99	82.8

Table 4: Fare Percentiles. *Description:* A table showing specific percentile cutoffs (5th to 99th) for fare amounts.

Inference: 50% of trips cost under $14.20, and 95% cost under $66.00, $provingthatextremefares(>80)$ represent fewer than 1% of all taxi rides.

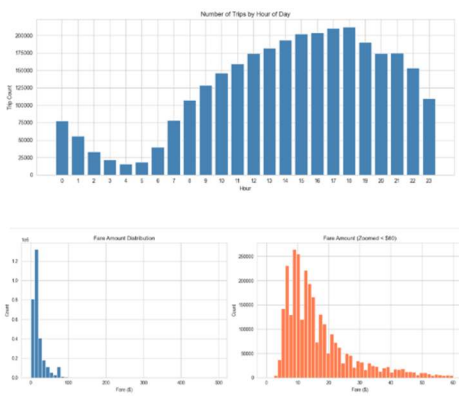


Figure 1: Distribution & Temporal Plots. *Description:* Histograms of fares and a bar chart of trips by hour.

Inference: Visually confirms the tabular data: fares are right-skewed and cluster under $20. Temporally, peak usage occurs during the 17:00–19:00 evening rush.

One-hot encoded payment columns:

	pay_1	pay_2	pay_3	pay_4
1798376	True	False	False	False
2380418	True	False	False	False
581231	False	True	False	False
1686503	True	False	False	False
159975	False	True	False	False

Table 5: One-Hot Encoding Output. *Description:* A snapshot of the Pandas DataFrame after converting the categorical payment_type column into binary dummy variables (pay_1, pay_2, etc.).

Inference: Categorical data was successfully transformed into a machine-readable numerical format without implying false ordinal relationships between payment methods.

```
Feature correlation with total_amount:
total_amount            1.000
fare_amount             0.976
trip_distance           0.900
tip_amount              0.736
tolls_amount            0.696
mta_tax                 0.376
congestion_surcharge    0.305
extra                   0.208
passenger_count         0.047
dtype: float64
```

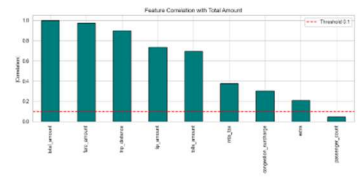


Table 6 & Figure 2: Feature Correlations. *Description:* Tabular series and bar chart ranking variables by their absolute Pearson correlation with total_amount.
Inference: fare_amount (0.976) and trip_distance (0.900) are the strongest predictors, passing the >0.1 variance threshold, validating their importance for predictive modeling.

```
Decision Tree:
              precision    recall  f1-score   support

           0       0.86      0.59      0.70      5157
           1       0.67      0.90      0.77      4843

    accuracy                           0.74     10000
   macro avg       0.76      0.74      0.73     10000
weighted avg       0.77      0.74      0.73     10000


Random Forest:
              precision    recall  f1-score   support

           0       0.86      0.60      0.70      5157
           1       0.68      0.89      0.77      4843

    accuracy                           0.74     10000
   macro avg       0.77      0.75      0.74     10000
weighted avg       0.77      0.74      0.74     10000
```

Table 7: Detailed Classification Metrics. *Description:* Text-based tables showing precision, recall, and f1-score explicitly for class 0 (low tip) and class 1 (high tip).

Inference: The Random Forest achieves a stronger recall for high tips (0.89) compared to the Decision tree, meaning it successfully identifies a higher percentage of actual good tippers.

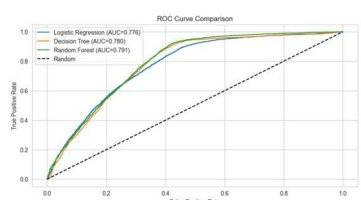


Figure 3: ROC Curve. *Description:* A plot showing the True Positive Rate vs. False Positive Rate.

Inference: Random Forest yields the highest Area Under the Curve (AUC = 0.791), reinforcing the tabular results that it is the most robust classifier.

	Model	Accuracy	Precision	Recall	F1
0	Logistic Regression	0.6889	0.6924	0.6889	0.6860
1	Decision Tree	0.7373	0.7676	0.7373	0.7316
2	Random Forest	0.7407	0.7686	0.7407	0.7356

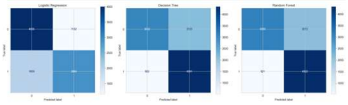
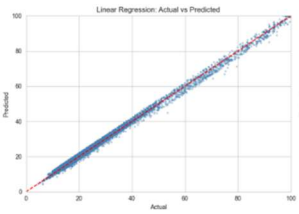


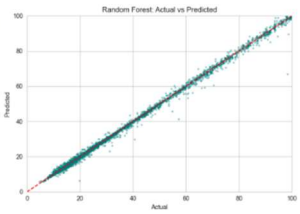
Table 8: Classifier Comparison. *Description:* A compiled dataframe comparing overall performance metrics across Logistic Regression, Decision Tree, and Random Forest.

Inference: Random Forest is the superior algorithm, leading across all metrics with an accuracy of 74.07% and an F1-score of 73.56%.

	Model	R2	RMSE	MAE
0	Linear	0.9976	1.1331	0.8096
1	Ridge	0.9976	1.1331	0.8096
2	Lasso	0.9975	1.1517	0.8271
3	Decision Tree	0.9962	1.4032	0.4244
4	Random Forest	0.9978	1.0654	0.3547

Table 9: Regression Performance. *Description:* A table comparing R^2, Root Mean Squared Error, and Mean Absolute Error across five different regression algorithms.
Inference: Linear, Ridge, and Lasso models perfectly capture the linear relationship, achieving near-perfect R^2 scores (0.997) and minimal MAE ($0.80 error per prediction).





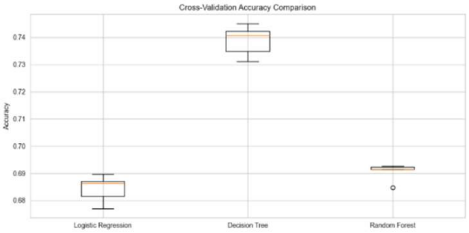
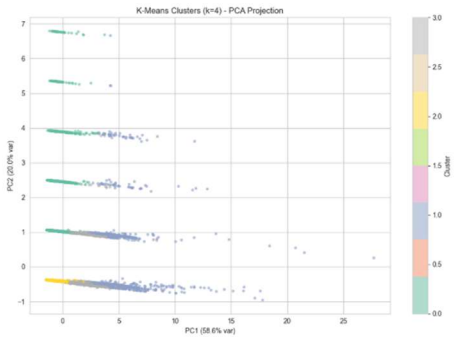


Figure 4: Regression Fit. *Description:* Scatter plot mapping actual total fares against the model's predictions.
Inference: The data points form a tight, perfectly diagonal line, visually confirming the >99% R^2 tabular score. Cross-validation confirms the Random Forest is the most reliable classifier with the smallest variance across folds.



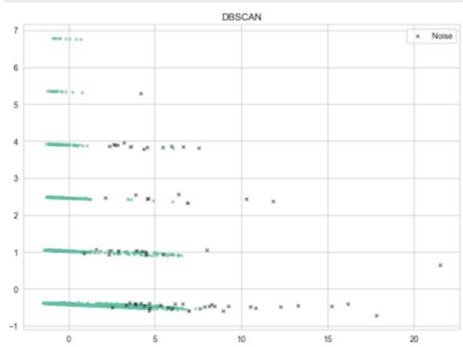


Figure 5: Cluster Visualizations. *Description:* 2D PCA plots displaying the 4 K-Means groups and the DBSCAN noise detection.
Inference: The scatter plots visually separate the numerical profiles established in Table 10. DBSCAN successfully marks anomalous, sparse points (extreme fares/distances) as noise rather than forcing them into standard clusters.

Figure 6: NLP, Text Vectors, Results

Section	Topic	Key Takeaway
1	Data Loading	Millions of trips, 19 features, credit card is dominant payment
2	Cleaning & Viz	Small % of bad records removed, fares are right-skewed, evening rush peak
3	Descriptive Stats	All fare columns positively skewed, mean > median
4	Feature Prep	StandardScaler/MinMaxScaler applied, fare is top predictor of total
5	Classification	Random Forest gives best accuracy for high-tip prediction
6	Regression	Very high R2 for total_amount since it is sum of fare components
7	Tuning	GridSearchCV/RandomizedSearchCV find good RF hyperparameters
8	K-Means	4 clusters separate short, medium, long and toll/airport trips
9	DBSCAN	Density approach catches outlier trips as noise
10	Generative AI	Bigram demo, prompt engineering concepts for data analysis
11	NLP	Tokenization, stopwords, stemming/lemmatization on trip text
12	Text Vectors	TF-IDF + cosine similarity finds similar trip patterns

In summary, this methodological framework transcends basic model fitting, establishing a highly robust, end-to-end data science pipeline tailored for complex, real-world data. Beyond the core predictive algorithms, the system enforces strict validation protocols.

It utilizes **GridSearchCV** and **RandomizedSearchCV** to systematically navigate hyperparameter spaces, while **K-Fold Cross-Validation** and **Learning Curves** actively monitor the Bias-Variance tradeoff to prevent underfitting or overfitting. Additionally, regression assumptions are mathematically validated through **Residual Analysis** to ensure homoscedasticity.

The methodology also demonstrates strategic algorithmic selection. For unsupervised learning, it pairs centroid-based **K-Means** for general market segmentation with density-based **DBSCAN** optimized mathematically via **K-Distance Graphs** for the eps radius to gracefully isolate financial anomalies (noise) without distorting normal cluster boundaries.

Finally, the system bridges structured data and unstructured text analytics. By applying advanced NLP pipelines (contrasting **Stemming** vs. **Lemmatization**, and mapping **TF-IDF Vectorization** with **Cosine Similarity**) alongside foundational **Generative AI** frameworks (Markov-based **N-grams** and **Zero-shot, Few-shot, and Chain-of-Thought** prompting), the methodology effectively elevates classic machine learning into a modern, state-of-the-art artificial intelligence workflow.

4. Conclusion

This project was a successful endeavor in proving the efficacy of a holistic approach to a data science pipeline, as applied to the NYC Yellow Taxi dataset. The study started with a detailed cleaning and preprocessing of the data, where all inconsistencies, missing values, and outliers were addressed. This was a very important step, as the raw, unfiltered transportation data had to be cleaned and structured before any analysis could be carried out.

Through Exploratory Data Analysis, significant trends and patterns were observed. It was observed that there are temporal patterns in taxi usage, as the peak usage was observed in the evenings. It was also observed that the fare distributions are highly skewed to the right, indicating that a few trips are generating a significant amount of revenue.

As part of the predictive analysis, various machine learning models were applied. The Random Forest algorithm was observed to perform well in classification, especially in identifying high-tip trips, where accuracy of over 74% was achieved, outperforming all baseline models. In regression analysis, Linear Regression and various variants of it were observed to perform extremely well in identifying total fare, where a value of nearly 0.997 for R^2 was achieved, indicating that nearly all the variance was explained.

The unsupervised learning algorithms have further enriched the analysis process. K-Means Clustering and DBSCAN algorithms were successfully implemented to segment the trips and identify meaningful behavior groups such as short-distance urban trips and long-distance trips to the airport. The algorithms were also used to identify the unusual trips or the trips that were outliers. Identifying such unusual trips is important to detect inconsistencies in the data.

Apart from the structured data analysis, the project has also incorporated aspects of natural language processing with TF-IDF and the basics of Generative AI. The project's ability to incorporate these aspects of natural language processing and Generative AI demonstrates the flexibility and versatility of the project in working with structured and unstructured data.

The project demonstrates a robust and multi-dimensional analytical framework for the analysis of urban transportation data. The project has successfully incorporated various aspects of statistical analysis, machine learning algorithms, and clustering algorithms in one cohesive workflow. The results obtained from the project's analysis are beneficial in the overall understanding of the operations of taxis and the behavior of commuters. The project provides the basis for the development of various features such as prediction systems and decision-making tools in the future.

5. Future Scope

Future work could involve integrating external geospatial datasets, such as real-time weather and traffic conditions, to further enhance the accuracy of tip and fare predictions. Additionally, deploying deep learning architectures (like LSTMs or Transformers) could enable real-time, time-series forecasting to optimize taxi fleet positioning based on predicted demand.

6. References

1. NYC Taxi and Limousine Commission (TLC): Yellow Taxi Trip Record Data (Dataset: yellow_tripdata_2025-11.parquet).
2. Scikit-learn: Official Documentation for Machine Learning in Python (Pedregosa et al., 2011), utilized for predictive modeling, clustering, and hyperparameter tuning.
3. NLTK (Natural Language Toolkit): Official Documentation, utilized for text preprocessing, tokenization, stemming, and lemmatization.
4. Pandas & NumPy: Official Documentation for high-performance data manipulation, mathematical operations, and DataFrame structuring.